EI331 Signals and Systems

Lecture 15

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April 16, 2019

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1. Recap

2. Properties of CT Fourier Transform

Recap: Fourier Transform for $L_1(\mathbb{R})$ Functions

Fourier transform and its inverse as principal values

$$\mathcal{F}\{x\}(j\omega) = \lim_{T\to\infty} \int_{-T}^{T} x(t)e^{-j\omega t}dt$$

$$\mathcal{F}^{-1}(X)(t) = \lim_{W\to\infty} \frac{1}{2\pi} \int_{-W}^{W} X(j\omega)e^{j\omega t}d\omega$$

Theorem. (Pointwise inversion)

1. If $x \in L_1(\mathbb{R})$ satisfies Dirichlet conditions on all finite intervals,

$$(\mathcal{F}^{-1} \circ \mathcal{F}\{x\})(t) = \frac{x(t_+) + x(t_-)}{2}, \quad \forall t$$

2. If $X \in L_1(\mathbb{R})$ satisfies Dirichlet conditions on all finite intervals,

$$(\mathcal{F} \circ \mathcal{F}^{-1}\{X\})(j\omega) = \frac{X(j\omega_+) + X(j\omega_-)}{2}, \quad \forall \omega$$

3. If $x \in C(\mathbb{R}) \cap L_1(\mathbb{R})$ and $\mathcal{F}\{x\} \in L_1(\mathbb{R})$, then $\mathcal{F}^{-1} \circ \mathcal{F}\{x\} = x$

Recap: Fourier Transform for $L_1(\mathbb{R})$ Functions

Fourier transform pairs

case	$x(t)$	$X(j\omega)$
1	$e^{-at}u(t), \quad \text{Re}\, a > 0$	$\dfrac{1}{a + j\omega}$
3	$e^{-a\|t\|}, \quad \text{Re}\, a > 0$	$\dfrac{2a}{a^2 + \omega^2}$
3	$e^{-at^2}, \quad a > 0$	$\sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}$
1	$u(t+T) - u(t-T)$	$\dfrac{2\sin(\omega T)}{\omega}$
2	$\dfrac{\sin(\omega_c t)}{\pi t}$	$u(\omega + \omega_c) - u(\omega - \omega_c)$

Recap: Fourier Transform for Generalized Functions

Schwarz space $\mathcal{S} = \mathcal{S}(\mathbb{R})$ of rapidly decreasing functions

$$\mathcal{S} = \{\phi \in C^\infty(\mathbb{R}) : \|\phi\|_{\ell,k} < \infty, \forall \ell, k \in \mathbb{N}\}$$

where $\|\phi\|_{\ell,k} = \sup_{t\in\mathbb{R}} |t^\ell \phi^{(k)}(t)|$. Note $\mathcal{F}\{\mathcal{S}\} = \mathcal{S}$.

Fourier transform of tempered distributions

$$(x, \phi) \triangleq \int_{\mathbb{R}} x(\xi)\phi(\xi)d\xi$$

- If $x_n \to x$ in distribution, then $\mathcal{F}\{x\} \triangleq \lim_n \mathcal{F}\{x_n\}$ in distribution

$$(x, \phi) = \lim_n (x_n, \phi) \implies (\mathcal{F}\{x\}, \phi) \triangleq \lim_n (\mathcal{F}\{x_n\}, \phi), \quad \forall \phi \in \mathcal{S}$$

- Alternatively, $\quad (\mathcal{F}\{x\}, \phi) \triangleq (x, \mathcal{F}\{\phi\}), \quad \forall \phi \in \mathcal{S}$

Recap: Fourier Transform for Generalized Functions

Fourier transform pairs

$x(t)$	$X(j\omega)$
$\delta(t)$	1
$\delta(t-t_0)$	$e^{-j\omega t_0}$
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$

$$\int_{\mathbb{R}} \delta(t-t_0)e^{-j\omega t}dt = e^{-j\omega t_0}, \qquad \int_{\mathbb{R}} e^{j(\omega_1-\omega_2)t}dt = 2\pi\delta(\omega_1-\omega_2)$$

$$\int_{\mathbb{R}} X(j\omega)e^{j\omega t}d\omega = \int_{\mathbb{R}}\int_{\mathbb{R}} x(\tau)e^{j\omega(t-\tau)}d\tau d\omega = \int_{\mathbb{R}} x(\tau)2\pi\delta(t-\tau)d\tau = 2\pi x(t)$$

Recap: Fourier Transform for Periodic Functions

Fourier series

$$x(t) = \sum_{k=-\infty}^{\infty} \hat{x}[k] e^{jk\omega_0 t}$$

Fourier transform

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi \hat{x}[k] \delta(\omega - k\omega_0)$$

Equivalent representations

- $\hat{x}[k]$ is amplitude of component at $\omega_k = k\omega_0$
- $\hat{x}[k]\delta(\omega - k\omega_0)$ is density of component at $\omega_k = k\omega_0$

Cf. discrete random variable x in probability

$$\mathbb{E}f(X) = \sum_n f(x_n) p_n = \int_{\mathbb{R}} f(x) p(x) dx \quad \text{with } p(x) = \sum_n p_n \delta(x - x_n)$$

Recap: Relation between FS and FT

- Aperiodic signal x with finite support of length $< T$
- Periodic extension x_T of x

$$x_T(t) = \sum_{k=-\infty}^{\infty} x(t - kT)$$

NB. In other words, x is one period of periodic signal x_T

$$x \xleftrightarrow{\mathcal{F}} X$$

$$x_T \xleftrightarrow{\mathcal{FS}} \hat{x}_T$$

$$x_T \xleftrightarrow{\mathcal{F}} X_T$$

$$\omega_0 = \frac{2\pi}{T}$$

$$\hat{x}_T[k] = \frac{1}{T} X(jk\omega_0)$$

$$X_T(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi \hat{x}_T[k] \delta(\omega - k\omega_0)$$

$$= \sum_{k=-\infty}^{\infty} \omega_0 X(jk\omega_0) \delta(\omega - k\omega_0)$$

Contents

1. Recap

2. Properties of CT Fourier Transform

Properties of CT Fourier Transform

Linearity

$$\mathcal{F}\{ax + by\} = a\mathcal{F}\{x\} + b\mathcal{F}\{y\}$$

Time shifting

$$\mathcal{F}\{\tau_{t_0} x\} = E_{-t_0}\mathcal{F}\{x\} \qquad \text{or} \qquad x(t - t_0) \xleftrightarrow{\mathcal{F}} e^{-j\omega t_0} X(j\omega)$$

where $(E_a f)(s) = e^{jas} f(s)$

Proof.

$$\int_{\mathbb{R}} x(t - t_0) e^{-j\omega t} dt = \int_{\mathbb{R}} x(s) e^{-j\omega(s+t_0)} ds = e^{-j\omega t_0} \int_{\mathbb{R}} x(s) e^{-j\omega s} ds$$

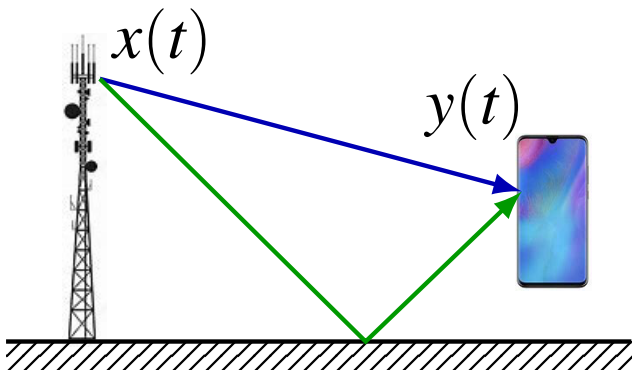
Frequency shifting

$$\mathcal{F}\{E_{\omega_0} x\} = \tau_{\omega_0} \hat{x} \qquad \text{or} \qquad e^{j\omega_0 t} x(t) \xleftrightarrow{\mathcal{F}} X(j(\omega - \omega_0))$$

Example: Multipath Effect

Transmitter: $x(t)$

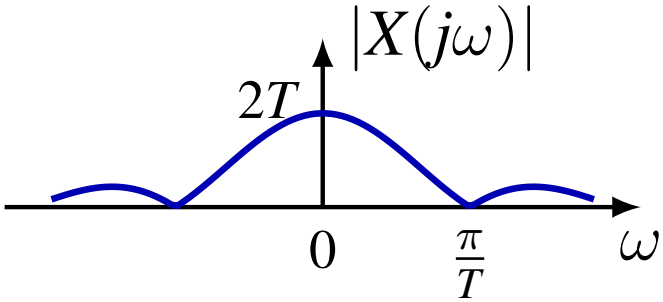
Receiver: $y(t) = a_0 x(t) + a_1 x(t-\tau)$



$$Y(j\omega) = a_0 X(j\omega) + a_1 e^{-j\omega\tau} X(j\omega)$$
$$= (a_0 + a_1 e^{-j\omega\tau}) X(j\omega)$$

Let $x(t) = u(t+T) - u(t-T)$.

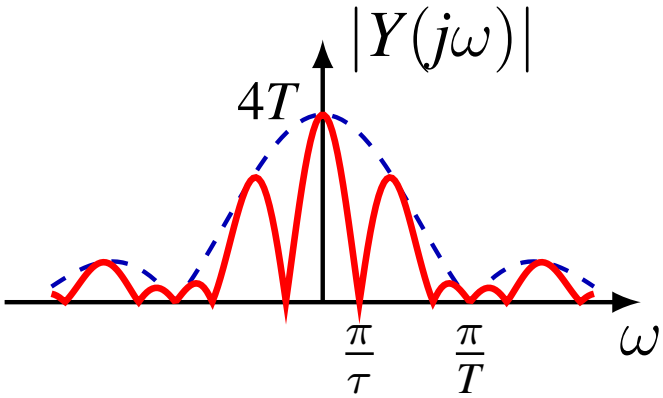
$$X(j\omega) = \frac{2\sin(\omega T)}{\omega}$$



$$Y(j\omega) = (a_0 + a_1 e^{-j\omega\tau}) \frac{2\sin(\omega T)}{\omega}$$

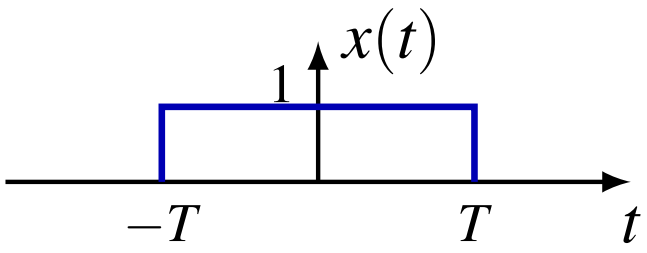
For $a_0 = a_1 = 1$,

$$Y(j\omega) = 4e^{-j\omega\tau/2} \cos(\frac{\omega\tau}{2}) \frac{\sin(\omega T)}{\omega}$$



Example: Modulation

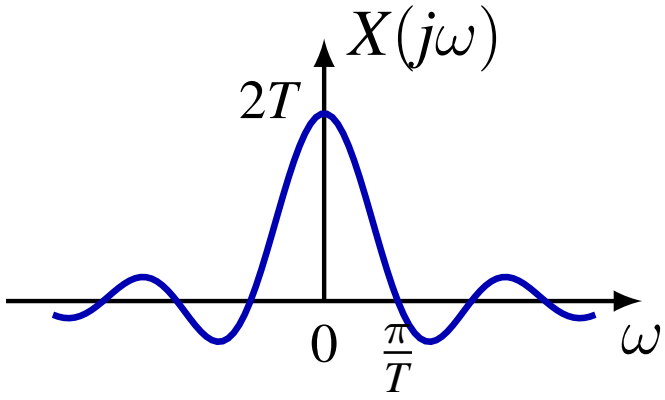
Baseband signal: $x(t)$



Modulated signal

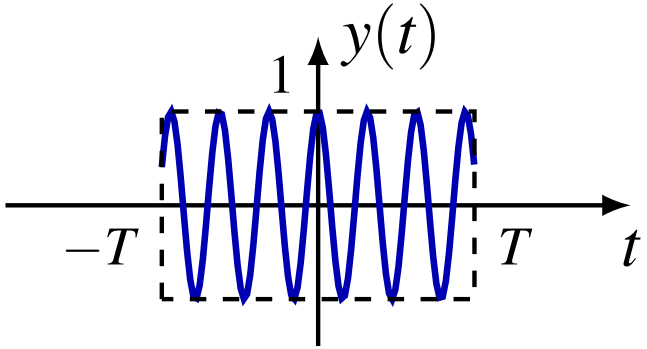
$$y(t) = x(t)\cos(\omega_c t) = \frac{1}{2}x(t)(e^{j\omega_c t} + e^{-j\omega_c t})$$

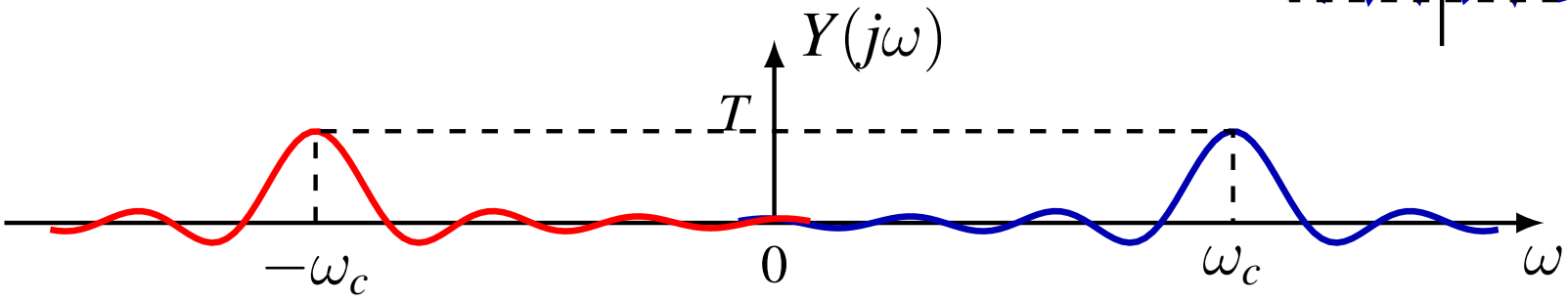
$$Y(j\omega) = \frac{1}{2}X(j(\omega - \omega_c)) + \frac{1}{2}X(j(\omega + \omega_c))$$



For $x(t) = u(t+T) - u(t-T)$,

$$Y(j\omega) = \frac{\sin((\omega - \omega_c)T)}{\omega - \omega_c} + \frac{\sin((\omega + \omega_c)T)}{\omega + \omega_c}$$





Properties of CT Fourier Transform

Time reversal

$$\mathcal{F}\{Rx\} = R\mathcal{F}\{x\} \qquad \text{or} \qquad x(-t) \xleftrightarrow{\mathcal{F}} X(-j\omega)$$

Conjugation

$$\mathcal{F}\{x^*\} = R\overline{\mathcal{F}\{x\}} \qquad \text{or} \qquad x^*(t) \xleftrightarrow{\mathcal{F}} X^*(-j\omega)$$

Proof. $\mathcal{F}\{x^*\}(j\omega) = \int_{\mathbb{R}} x^*(t)e^{-j\omega t}dt = \left(\int_{\mathbb{R}} x(t)e^{j\omega t}dt\right)^* = X^*(-j\omega)$

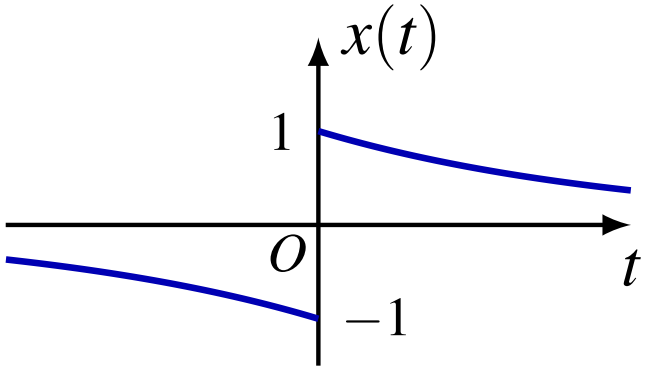
Symmetry

- x even $\iff$ X even, $\quad x$ odd $\iff$ X odd
- x real $\iff$ $X(-j\omega) = X^*(j\omega)$
- x real and even $\iff$ X real and even
- x real and odd $\iff$ X purely imaginary and odd

Example

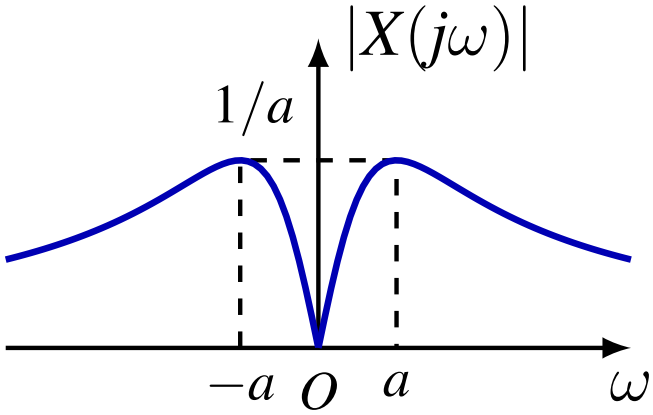
For $a > 0$,

$$x(t) = \begin{cases} e^{-at}, & t > 0 \\ -e^{at}, & t < 0 \end{cases}$$



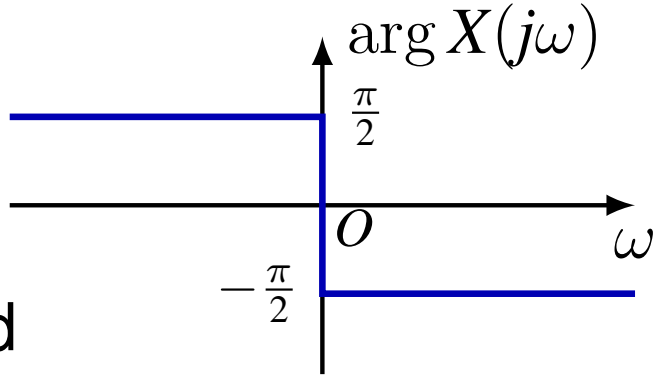
Let $y(t) = e^{-at}u(t)$, with

$$Y(j\omega) = \frac{1}{a + j\omega}$$



Since $x(t) = y(t) - y(-t)$.

$$X(j\omega) = Y(j\omega) - Y(-j\omega)$$
$$= \frac{1}{a + j\omega} - \frac{1}{a - j\omega} = \frac{-2j\omega}{a^2 + \omega^2}$$



x real & odd, X purely imaginary & odd

Time and Frequency Scaling

For $a \neq 0$,

$$\mathcal{F}\{S_a x\} = \frac{1}{|a|} S_{\frac{1}{a}} \mathcal{F}\{x\} \qquad \text{or} \qquad x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

Proof. Change variables by $\tau = at$.

For $a > 0$,

$$\int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(\tau) e^{-j\frac{\omega}{a}\tau} \frac{d\tau}{a} = \frac{1}{a} X\left(\frac{j\omega}{a}\right)$$

For $a < 0$,

$$\int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt = \int_{\infty}^{-\infty} x(\tau) e^{-j\frac{\omega}{a}\tau} \frac{d\tau}{a} = -\frac{1}{a} X\left(\frac{j\omega}{a}\right)$$

Time and Frequency Scaling

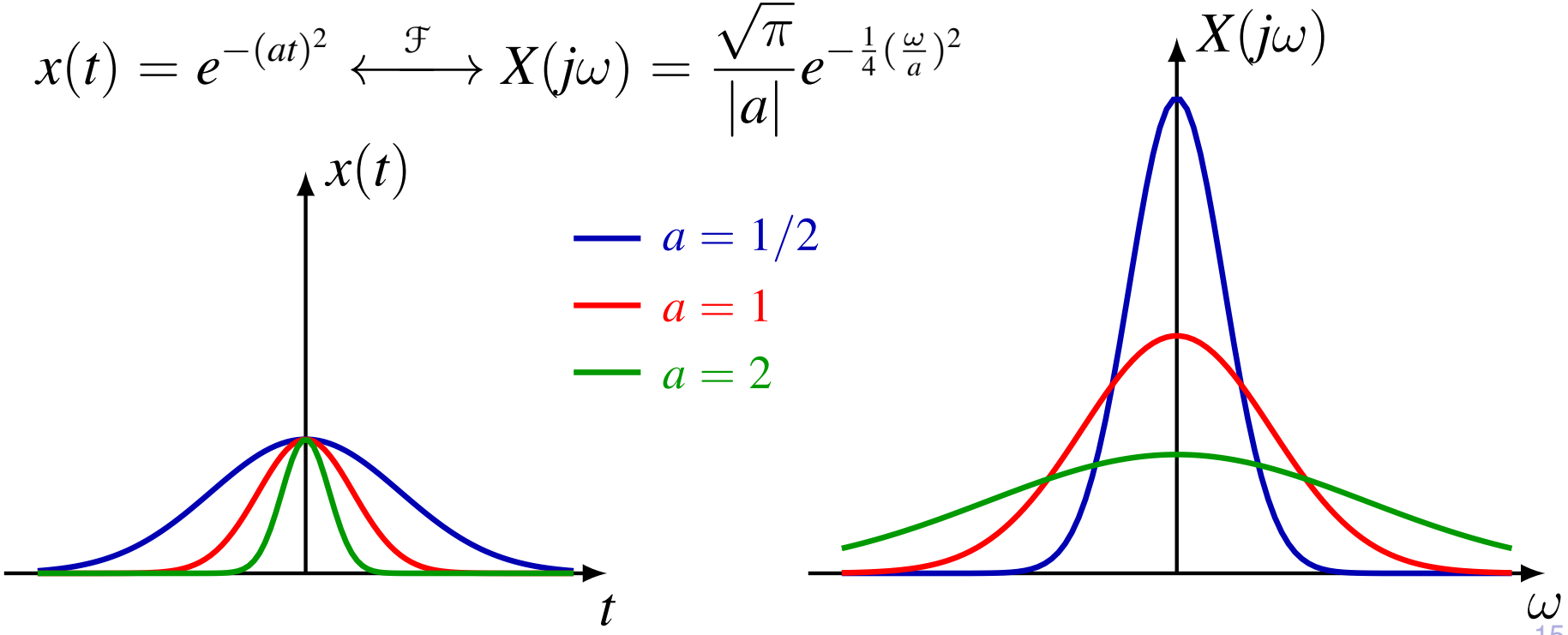
$$\mathcal{F}\{S_a x\} = \frac{1}{|a|} S_{\frac{1}{a}} \mathcal{F}\{x\} \qquad \text{or} \qquad x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

compression in time $\Longleftrightarrow$ stretching in frequency

stretching in time $\Longleftrightarrow$ compression in frequency

e.g. audio: faster playback $\Longrightarrow$ higher pitch

$$x(t) = e^{-(at)^2} \xleftrightarrow{\mathcal{F}} X(j\omega) = \frac{\sqrt{\pi}}{|a|} e^{-\frac{1}{4}(\frac{\omega}{a})^2}$$



Differentiation

$$x'(t) \xleftrightarrow{\mathcal{F}} j\omega X(j\omega), \quad x^{(k)}(t) \xleftrightarrow{\mathcal{F}} (j\omega)^k X(j\omega)$$

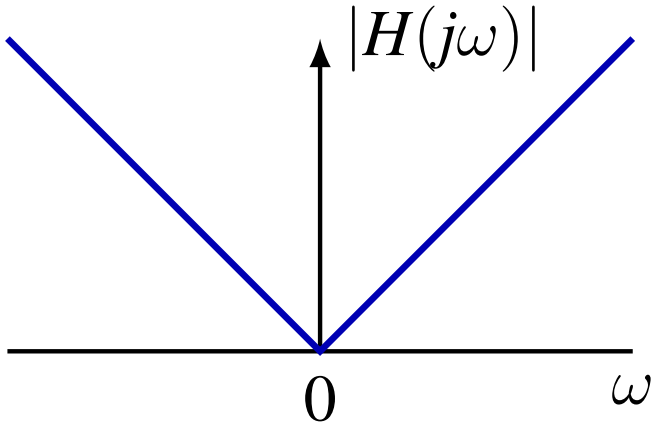
Proof. Differentiate under integral sign

$$x(t) = \frac{1}{2\pi}\int_{\mathbb{R}} X(j\omega)e^{j\omega t}d\omega \implies x'(t) = \frac{1}{2\pi}\int_{\mathbb{R}} j\omega X(j\omega)e^{j\omega t}d\omega$$

$y = x'$ is output of differentiator with frequency response

$$H(j\omega) = \mathcal{F}\{\delta'\}(j\omega) = \int_{\mathbb{R}} \delta'(t)e^{-j\omega t}dt = -\frac{d}{dt}e^{-j\omega t}\Big|_0 = j\omega$$

- $Y(j\omega) = H(j\omega)X(j\omega)$
- amplifies high frequencies
- suppresses low frequencies
- DC completely eliminated



Integration

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau \xleftrightarrow{\mathcal{F}} Y(j\omega) = \frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$$

Since $x(t) = y'(t)$, by differentiation property

$$X(j\omega) = j\omega Y(j\omega) \implies Y(j\omega) = \frac{1}{j\omega}X(j\omega) \text{ for } \omega \neq 0$$

Intuitively, $y(t)$ has DC component

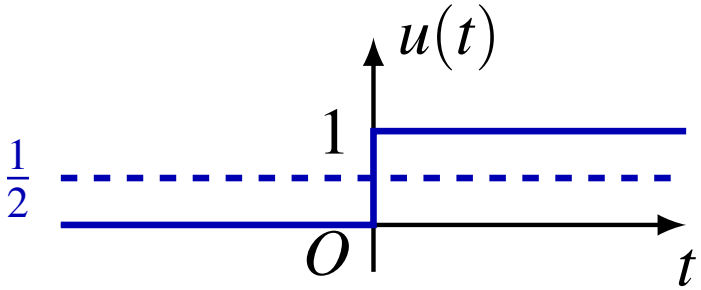
$$\begin{aligned}\bar{y} &= \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} y(t)dt = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T}\int_{\mathbb{R}} x(\tau)u(t-\tau)d\tau dt \\ &= \int_{\mathbb{R}} x(\tau)\left(\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} u(t-\tau)dt\right)d\tau = \frac{1}{2}X(0)\end{aligned}$$

So Y also has component $2\pi\bar{y}\delta(\omega) = \pi X(0)\delta(\omega)$

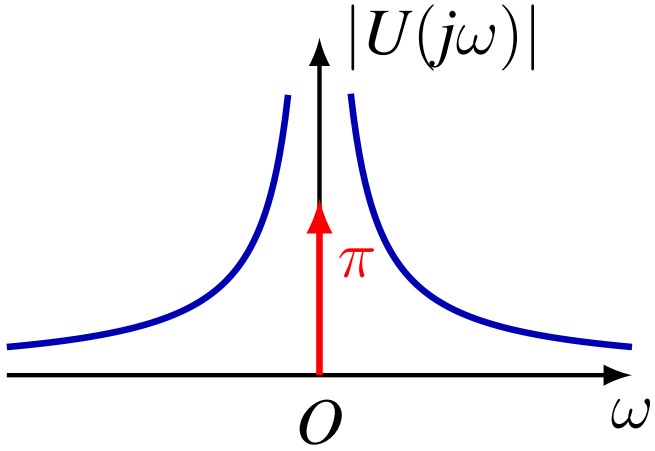
Integration

Example. Unit step function

$$u(t) = \int_{-\infty}^{t} \delta(\tau) d\tau$$



$$U(j\omega) = \frac{1}{j\omega}\mathcal{F}\{\delta\}(j\omega) + \pi\mathcal{F}\{\delta\}(0)\delta(\omega)$$
$$= \frac{1}{j\omega} + \pi\delta(\omega)$$

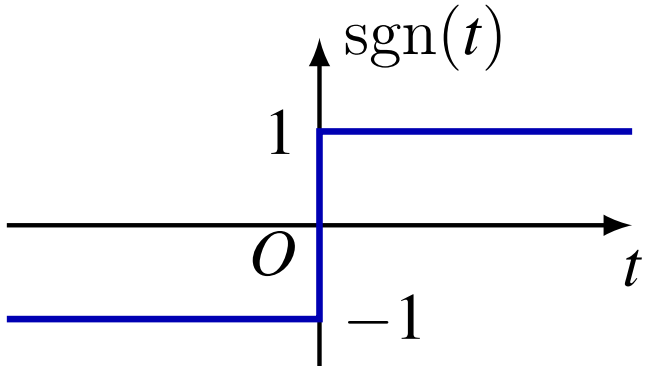


Example. Sign function

$$\mathrm{sgn}(t) = \begin{cases} 1, & t > 0 \\ -1 & t < 0 \end{cases}$$

Since $\mathrm{sgn} = 2u - 1$,

$$\mathcal{F}\{\mathrm{sgn}\}(j\omega) = 2U(j\omega) - 2\pi\delta(\omega) = \frac{2}{j\omega}$$



Integration

Example. Unit ramp function $u_{-2}(t) = tu(t)$

$$u_{-2}(t) = \int_{-\infty}^{t} u(\tau)d\tau$$

Integration property suggests

$$\mathcal{F}\{u_{-2}\}(j\omega) = \frac{1}{j\omega}U(j\omega) + \pi U(0)\delta(\omega) = -\frac{1}{\omega^2} + \frac{\pi}{j\omega}\delta(\omega) + \pi U(0)\delta(\omega)$$

But what's $U(0) = \frac{1}{j0} + \pi\delta(0)$? What's $\frac{\pi}{j\omega}\delta(\omega)$? **Not well-defined!**

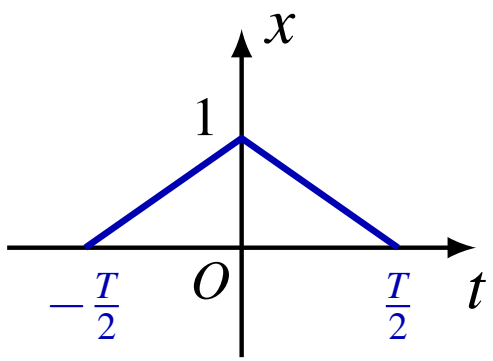
Integration property **not applicable** here!

Will see $\mathcal{F}\{u_{-2}\} = -\frac{1}{\omega^2} + j\pi\delta'(\omega)$

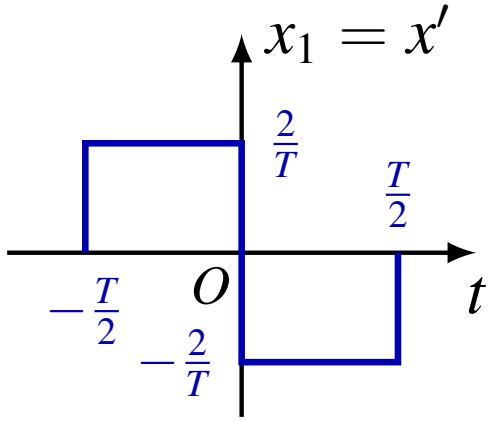
Rule of thumb. Applicable when formula is well-defined

Example

$$x(t) = \left(1 - \frac{2|t|}{T}\right)\left[u(t+\frac{T}{2}) - u(t-\frac{T}{2})\right]$$

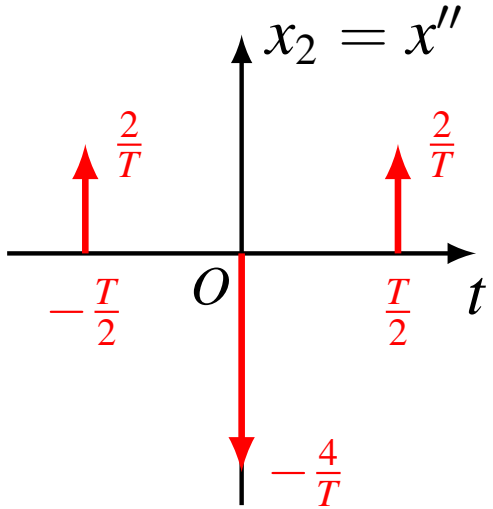


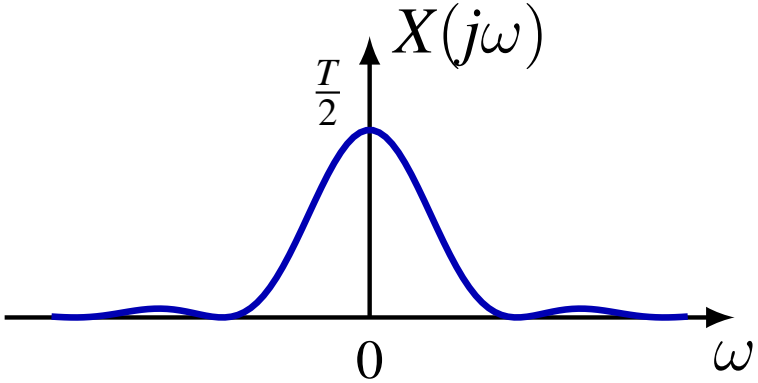
$$X_2(j\omega) = \frac{2}{T}[e^{j\frac{\omega T}{2}} - 2 + e^{-j\frac{\omega T}{2}}] = -\frac{8}{T}\sin^2\left(\frac{\omega T}{4}\right)$$



$$X_1(j\omega) = \frac{X_2(j\omega)}{j\omega} + \pi X_2(0)\delta(\omega) = -\frac{8\sin^2\left(\frac{\omega T}{4}\right)}{j\omega T}$$

$$X(j\omega) = \frac{X_1(j\omega)}{j\omega} + \pi X_1(0)\delta(\omega) = \frac{8\sin^2\left(\frac{\omega T}{4}\right)}{\omega^2 T}$$





Example: Sign Function sgn

$$x(t) = \mathrm{sgn}(t) \xleftrightarrow{\mathcal{F}} X(j\omega) = \frac{2}{j\omega}$$

- Neither x nor X is in $L_1(\mathbb{R})$, above interpreted in the sense of distribution, i.e.

$$(X, \phi) = (x, \mathcal{F}\{\phi\}), \quad \forall \phi \in \mathcal{S}$$

  or for approximating sequence $x_n \to x$,

$$(X, \phi) = \lim_n (X_n, \phi), \quad \forall \phi \in \mathcal{S}$$

- Since $\frac{2}{j\omega}$ not integrable at $\omega = 0$, X should be interpreted as principal value, often denoted $X(j\omega) = \mathrm{pv}\left(\frac{2}{j\omega}\right)$, i.e.

$$(X, \phi) = \int_{\mathbb{R}} \mathrm{pv}\left(\frac{2}{j\omega}\right) \phi(\omega) d\omega \triangleq \lim_{\epsilon \to 0} \int_{|\omega| \geq \epsilon} \frac{2}{j\omega} \phi(\omega) d\omega$$

Example: Sign Function sgn

Claim. Principal value below is well-defined

$$\int_{\mathbb{R}} \mathrm{pv}\left(\frac{1}{\omega}\right)\phi(\omega)d\omega \triangleq \lim_{\epsilon\to 0}\int_{|\omega|\geq\epsilon}\frac{1}{\omega}\phi(\omega)d\omega$$

Proof.

- $\int_{|\omega|\geq 1}\frac{1}{\omega}\phi(\omega)d\omega$ well-defined, since ϕ is rapidly decreasing
- Since $\int_{1\geq|\omega|\geq\epsilon}\frac{1}{\omega}d\omega = 0$,

$$\int_{1\geq|\omega|\geq\epsilon}\frac{1}{\omega}\phi(\omega)d\omega = \int_{1\geq|\omega|\geq\epsilon}\frac{\phi(\omega)-\phi(0)}{\omega}d\omega$$

- By Intermediate Value Theorem, $\left|\frac{\phi(\omega)-\phi(0)}{\omega}\right| = |\phi'(\xi)| \leq \|\phi\|_{0,1}$

$$\left|\int_{1\geq|\omega|\geq\epsilon}\frac{1}{\omega}\phi(\omega)d\omega\right| = \int_{1\geq|\omega|\geq\epsilon}\left|\frac{\phi(\omega)-\phi(0)}{\omega}\right|d\omega \leq 2\|\phi\|_{0,1}$$

Example: Sign Function sgn

Claim.

$$\mathcal{F}\{\mathrm{sgn}\}(j\omega) = \mathrm{pv}\left(\frac{2}{j\omega}\right)$$

Proof. Let $x_a(t) = \mathrm{sgn}(t)e^{-a|t|}$ for $a > 0$. $x_a \to x$ pointwise and in **distribution** as $a \to 0$ (**legal** to push limit inside integral)

$$\lim_{a\to 0}\int_{\mathbb{R}} x_a(t)\phi(t) = \int_{\mathbb{R}} \lim_{a\to 0} x_a(t)\phi(t) = \int_{\mathbb{R}} x(t)\phi(t)dt$$

Recall

$$X_a(j\omega) = \frac{-2j\omega}{a^2+\omega^2}$$

For $\omega \neq 0$, $X_a(j\omega) \to \frac{2}{j\omega}$ pointwise as $a \to 0$. But need

$$\int_{\mathbb{R}} X_a(j\omega)\phi(\omega)d\omega \to \int_{\mathbb{R}} \mathrm{pv}\left(\frac{2}{j\omega}\right)\phi(\omega)d\omega$$

Example: Sign Function sgn

Claim.

$$\lim_{a\to 0}\int_{\mathbb{R}}\frac{\omega}{a^2+\omega^2}\phi(\omega)d\omega=\lim_{\epsilon\to 0}\int_{|\omega|\geq\epsilon}\frac{1}{\omega}\phi(\omega)d\omega$$

Proof. Since $\frac{\omega}{a^2+\omega^2}\phi(\omega)\in L_1(\mathbb{R})$,

$$\int_{\mathbb{R}}\frac{\omega}{a^2+\omega^2}\phi(\omega)d\omega=\lim_{\epsilon\to 0}\int_{|\omega|\geq\epsilon}\frac{\omega}{a^2+\omega^2}\phi(\omega)d\omega$$

Need to show $\lim_{a\to 0}\lim_{\epsilon\to 0}|J|=0$, where

$$J=\int_{|\omega|\geq\epsilon}\left(\frac{\omega}{a^2+\omega^2}-\frac{1}{\omega}\right)\phi(\omega)d\omega=\int_{|\omega|\geq\epsilon}\frac{-a^2}{\omega(a^2+\omega^2)}\phi(\omega)d\omega$$

Example: Sign Function sgn

Proof (cont'd). Using $0 = \int_{|\omega|\geq\epsilon} \frac{-a^2}{\omega(a^2+\omega^2)} d\omega$, we obtain

$$J = \int_{|\omega|\geq\epsilon} \frac{-a^2}{a^2+\omega^2} \cdot \frac{\phi(\omega)-\phi(0)}{\omega} d\omega.$$

By Intermediate Value Theorem, $|\frac{\phi(\omega)-\phi(0)}{\omega}| = |\phi'(\xi)| \leq \|\phi\|_{0,1}$

$$|J| \leq \int_{|\omega|\geq\epsilon} \left| \frac{-a^2}{a^2+\omega^2} \cdot \frac{\phi(\omega)-\phi(0)}{\omega} \right| d\omega \leq \int_{|\omega|\geq\epsilon} \frac{a^2}{a^2+\omega^2} \|\phi\|_{0,1} d\omega$$
$$\leq \int_{\mathbb{R}} \frac{a^2}{a^2+\omega^2} \|\phi\|_{0,1} d\omega = a\pi \|\phi\|_{0,1}$$

Therefore,

$$\lim_{a\to 0} \lim_{\epsilon\to 0} |J| = 0$$

Example: Unit Step Function

$$u(t) \xleftrightarrow{\mathcal{F}} U(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

NB. More precisely, $U(j\omega) = \mathrm{pv}\left(\frac{1}{j\omega}\right) + \pi\delta(\omega)$

Proof. Note $u(t) = \frac{1}{2}\,\mathrm{sgn}(t) + \frac{1}{2}$. By linearity

$$U(j\omega) = \frac{1}{2}\mathcal{F}\{\mathrm{sgn}\}(j\omega) + \frac{1}{2}\mathcal{F}\{1\}(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

Remark. Can use approximation approach but with caution.

- Let $x_a(t) = e^{-at}u(t)$ for $a > 0$. Note $x_a \to u$ pointwise and **in distribution** as $a \to 0$.
- For $\omega \neq 0$, $X_a(j\omega) = \frac{1}{a+j\omega} \to \frac{1}{j\omega}$ pointwise as $a \downarrow 0$.
- Tempting to conclude $U(j\omega) = \frac{1}{j\omega}$. ↯
- But need convergence **in distribution !**

Duality

$$\mathcal{F}\{f\}(\eta) = \int_{\mathbb{R}} f(\xi) e^{-j\eta\xi} d\xi$$

$$\mathcal{F}^{-1}\{f\}(\eta) = \frac{1}{2\pi} \int_{\mathbb{R}} f(\xi) e^{j\eta\xi} d\xi$$

Identical except for

- different signs in exponent of complex exponential
- constant factor $\frac{1}{2\pi}$

$$\mathcal{F}\{f\}(\eta) = \int_{\mathbb{R}} f(\xi) e^{-j\eta\xi} d\xi = \int_{\mathbb{R}} f(-\xi) e^{j\eta\xi} d\xi = \mathcal{F}^{-1}\{2\pi Rf\}(\eta)$$

Duality

$$f \xleftrightarrow{\mathcal{F}} g \iff g \xleftrightarrow{\mathcal{F}} 2\pi Rf \iff Rg \xleftrightarrow{\mathcal{F}} 2\pi f$$

Duality

$$\mathcal{F}\{f\}(\eta) = \int_{\mathbb{R}} f(\xi) e^{-j\eta\xi} d\xi$$

$$\mathcal{F}^{-1}\{f\}(\eta) = \frac{1}{2\pi} \int_{\mathbb{R}} f(\xi) e^{j\eta\xi} d\xi$$

Identical except for

- different signs in exponent of complex exponential
- constant factor $\frac{1}{2\pi}$

$$\overline{\mathcal{F}\{f\}} = \left(\int_{\mathbb{R}} f(\xi) e^{-j\eta\xi} d\xi \right)^* = \int_{\mathbb{R}} f^*(\xi) e^{j\eta\xi} d\xi = \mathcal{F}^{-1}\{2\pi f^*\}$$

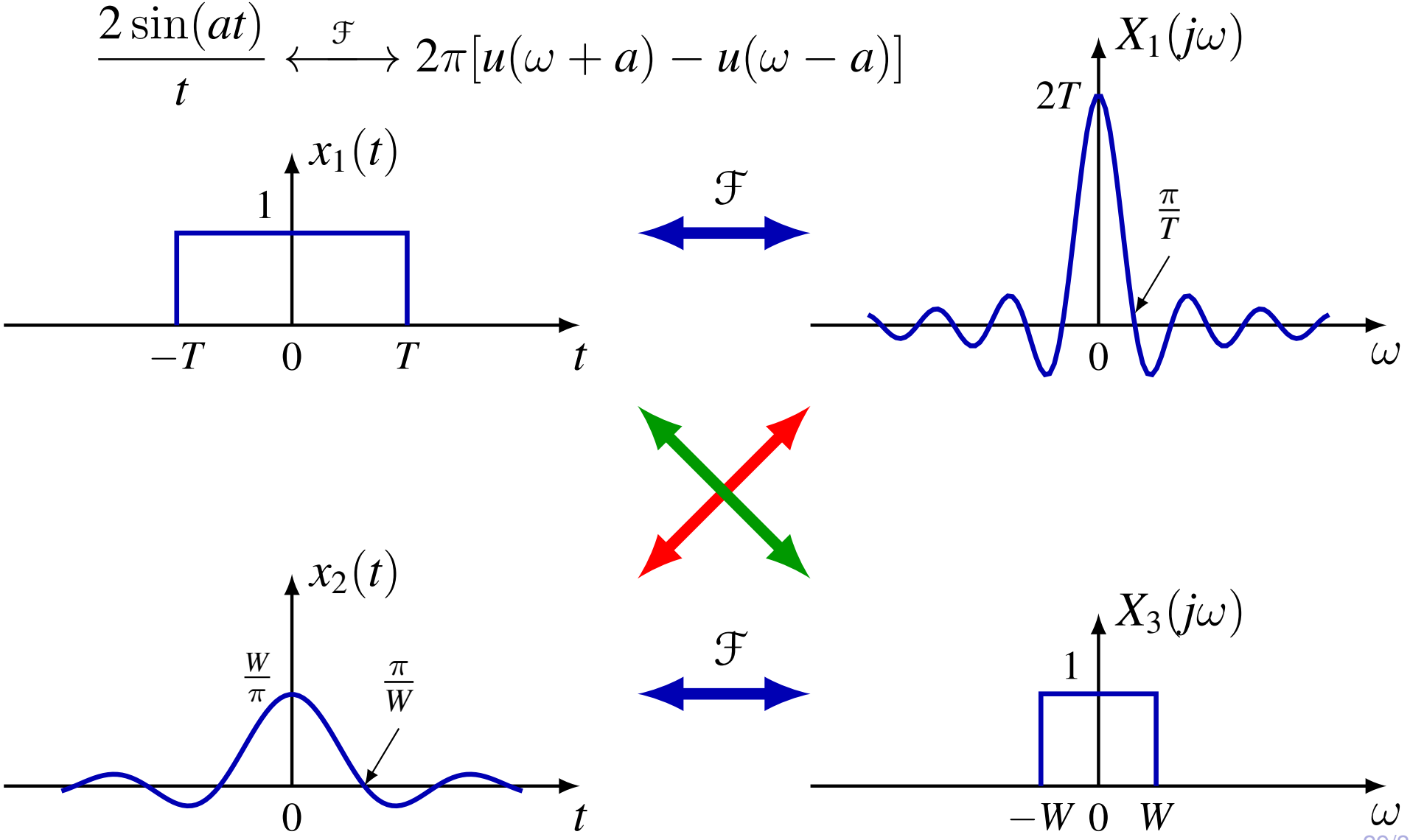
Duality

$$f \xleftrightarrow{\mathcal{F}} g \quad \Longleftrightarrow \quad g^* \xleftrightarrow{\mathcal{F}} 2\pi f^*$$

Duality

$$u(t+a) - u(t-a) \xleftrightarrow{\mathcal{F}} \frac{2\sin(a\omega)}{\omega}$$

$$\frac{2\sin(at)}{t} \xleftrightarrow{\mathcal{F}} 2\pi[u(\omega+a) - u(\omega-a)]$$



Duality

Example. $x(t) = \frac{1}{t}$ (more precisely, $\mathrm{pv}\left(\frac{1}{t}\right)$)

Know

$$\mathrm{sgn}(t) \xleftrightarrow{\mathcal{F}} \frac{1}{j\omega}$$

By duality

$$-\frac{1}{jt} \xleftrightarrow{\mathcal{F}} 2\pi\,\mathrm{sgn}(\omega)$$

- switch LHS and RHS, switch ω and t
- do either of following (but **not** both)
  - take complex conjugate of both sides
  - substitute $\omega \to -\omega$ or $t \to -t$ (but **not** both)
- multiply RHS by 2π

By linearity

$$\frac{1}{t} \xleftrightarrow{\mathcal{F}} -j2\pi\,\mathrm{sgn}(\omega)$$

Duality

Example. $x(t) = \frac{1}{1+t^2}$

Know

$$e^{-|t|} \xleftrightarrow{\mathcal{F}} \frac{2}{1+\omega^2}$$

By duality

$$\frac{2}{1+t^2} \xleftrightarrow{\mathcal{F}} 2\pi e^{-|\omega|}$$

- switch LHS and RHS, switch ω and t
- do either of following (but **not** both)
  - take complex conjugate of both sides
  - substitute $\omega \to -\omega$ or $t \to -t$ (but **not** both)
- multiply RHS by 2π

By linearity

$$\frac{1}{1+t^2} \xleftrightarrow{\mathcal{F}} \pi e^{-|\omega|}$$

Duality

Differentiation in frequency

$$-jtx(t) \xleftrightarrow{\mathcal{F}} \frac{dX(j\omega)}{d\omega}$$

Proof. Differentiate under integral sign

$$X(j\omega) = \int_{\mathbb{R}} x(t)e^{-j\omega t}dt \implies \frac{dX(j\omega)}{d\omega} = \int_{\mathbb{R}} (-jt)x(t)e^{-j\omega t}dt$$

Example. Unit ramp function $u_{-2} = tu(t)$

$$\mathcal{F}\{u_{-2}\} = j\mathcal{F}\{-jtu\} = j\frac{dU(j\omega)}{d\omega} = -\frac{1}{\omega^2} + j\pi\delta'(\omega)$$

Integration in frequency

$$-\frac{1}{jt}x(t) + \pi x(0)\delta(t) \xleftrightarrow{\mathcal{F}} \int_{-\infty}^{\omega} X(j\lambda)d\lambda$$